

DEGREE: BSc (Hons) Computer Science and Digitisation

Module: Statistics for Data Analysis

Assignment Title: Analytical and Predictive Statistics for Strategic Business Decision-Making

Assignment Type: Written assignment

Word Limit: 2000 words (+/- 200)

Weighting: 100%

Issue Date: 18/06/2026

Submission Date: 14/08/2026

Feedback Date: 28/08/2026

Plagiarism:

When submitting work for assessment, students should be aware of the InterActive/Canvas guidance and regulations in concerning plagiarism. All submissions should be your own, original work.

You must submit an electronic copy of your work. Your submission will be electronically checked.

Learner declaration		
Word count		
Use of proof-reader/proof-reading service (see guidance contained in Academic Misconduct Regulations on appropriate use)	YES	NO
If 'yes', please specify which service (e.g. Grammarly, Studiosity etc.)		
I confirm that I submit this work as my own work and that I have cited all sources I have used, and I understand that using sources without citing them correctly may be considered as Academic Misconduct.	YES	NO

I confirm that I have followed guidance on the acceptable use of AI tools for this assignment where such guidance has been issued by my tutors.	YES	NO
Where use of appropriately cited use of AI tools is permitted, I confirm that I have cited in accordance with the UCA Harvard Referencing Standard.	YES	NO

Harvard Referencing:

The Harvard Referencing System must be used. The Wikipedia, UKEssays.com or similar websites must **not** be used or referenced in your work.

By submitting this assessment, you are confirming that you have understood and are complying with the University's Academic Misconduct Regulations.

Plagiarism Detection (Turnitin)

BSBI uses Turnitin (an on-line text matching tool) to help identify cases of plagiarism. Turnitin will compare a submitted assignment with a database of journal articles, books, web pages and previously submitted assignments.

Signature (Student): _____ Date: _____

Introduction

Learning Outcomes:

LO1. Demonstrating an understanding and concepts of descriptive statistics which can be used to analyse the data better.

LO2. Demonstrating an understanding and concepts of inferential statistics which can be used to analyse the data better and infer the business value from this data.

LO3. Implementing statistical tests to implement inferential statistics and derive insights from a sample for the population parameters

Assessment Criteria: Weighting 100%
2000 words

Tasks

- a. You are provided with the Swiss data set. Begin your essay with an introduction discussing the significance of studying fertility rates and socioeconomic factors in understanding demographic trends.
- b. Identify and handle outliers or noise within the dataset using statistical methods.
- c. Use measures of central tendency and measures of dispersion to describe the distribution of variables such as fertility rates, education levels, and agricultural employment.
- d. Create visual representations, such as histograms or boxplots, to illustrate the distribution of key socioeconomic variables and fertility rates.
- e. Normalize the to bring variables such as fertility rates, education levels, and employment percentages to a comparable scale.
- f. Visualize the data using scatter plots, correlation matrices, or heatmaps to explore relationships between fertility rates and socioeconomic factors (e.g., education, income, agriculture).
- g. Conduct inferential statistical analysis to assess relationships and potential causality between fertility rates and socioeconomic indicators.
- h. Summarize the findings from the data analysis, focusing on the relationships between fertility and socioeconomic indicators.

GUIDANCE ON ASSESSMENT

All materials must be properly referenced under Harvard conventions. The length required is 2000 words with tasks equally weighted. The writing style should be formal academic / report writing style with in-text referencing to support your comments and observations. Originality, quality of argument and good structure are required. The written assignment should demonstrate sound understanding and ability to apply knowledge and theory of statistics. Additional marks being awarded for juxtaposition and insight of issues.

Grading Criteria

	Generic Criteria	90 - 100	80 - 89	70 - 79	60 - 69	50 - 59	40 - 49	30 - 39	0 - 29
Level 4	Knowledge of contexts, concepts, technologies and processes The extent to which knowledge is demonstrated: <i>relevant contextual or theoretical issues are identified, defined and described</i> <i>historical or contemporary practices are identified, defined and described</i> <i>appropriate technologies, methods and processes are identified, defined and described</i>	Outstanding breadth of knowledge of fundamental contextual and theoretical issues and critical concepts and their relationship to historical and contemporary practices Extensive knowledge of relevant and specialist technologies and processes	Extensive knowledge of fundamental contextual and theoretical issues and critical concepts and a widening appreciation of historical and contemporary practices In depth and broadening knowledge of appropriate technologies and processes	Significant knowledge of fundamental contextual and theoretical issues and critical concepts and a widening appreciation of historical and contemporary practices Significant knowledge of appropriate technologies and processes	Confident familiarity with fundamental contextual and theoretical issues and critical concepts Thorough knowledge of appropriate technologies and processes	Familiar with fundamental contextual and theoretical issues and critical concepts Sound knowledge of appropriate technologies and processes	Adequate knowledge of fundamental contextual and theoretical issues and critical concepts Adequate knowledge of appropriate technologies and processes	Limited knowledge of fundamental contextual and theoretical issues and critical concepts Limited knowledge of appropriate technologies and processes	Little or no knowledge of fundamental contextual and theoretical issues or critical concepts Little or no knowledge of appropriate technologies or processes

Understanding through application of knowledge The degree to which research methods are demonstrated: <i>relevant knowledge and information is compared, contrasted, manipulated, translated and interpreted</i> <i>knowledge and information is selected, analysed, synthesized and evaluated in order to generate creative ideas, practices, solutions, arguments or hypotheses</i>	Relevant knowledge is explored and interpreted when proposing solutions to projects and problems which demonstrate evidence of independent thought Outstanding ability to analyse and synthesise knowledge to produce own creative practice in standard situations and to evaluate results	Deep level of comprehension and exploration of relevant knowledge in seeking solutions to projects or problems Outstanding ability to analyse and synthesise knowledge in order to produce creative practice in standard situations and to evaluate the results	Deep level of comprehension of relevant knowledge in seeking solutions to projects or problems Strong ability to apply and analyse knowledge to produce creative practice in standard situations, with some evaluation of the results	Strong comprehension of relevant knowledge in seeking solutions to projects or problems Sound ability to apply and analyse knowledge to produce creative practice in standard situations	Sound comprehension of relevant knowledge in seeking solutions to projects or problems Sound ability to apply knowledge to produce creative practice in standard situations	Surface-level comprehension of relevant knowledge in seeking solutions to projects or problems Competent application of knowledge to the production of creative practice in standard situations	Incomplete comprehension of relevant knowledge in seeking solutions to projects or problems Limited ability to apply knowledge to produce creative practice in standard situations	Little or no comprehension of relevant knowledge in seeking solutions to projects or problems Little or no ability to apply relevant knowledge to produce creative practice in standard situations
Application of technical and professional skills The degree to which: <i>appropriate materials and media are selected, tested and utilised to realise and present ideas and solutions</i>	Accomplished and fluent application of appropriate practical and technical skills Outstanding application of appropriate transferable and	Relevant, accomplished and fluent application of basic practical and technical skills Outstanding application of fundamental	Relevant and accomplished application of basic practical and technical skills Highly effective application of fundamental transferable and	Strong application of basic practical and technical skills Strong application of fundamental transferable and	Sound application of basic practical and technical skills Sound application of fundamental transferable and	Competent application of practical and technical skills Competent application of fundamental transferable and	Rudimentary application of basic practical and technical skills Limited application of fundamental transferable and	Scant application of basic practical and technical skills Ineffective application of fundamental transferable and

<i>appropriate technologies, methods and processes are demonstrated</i> <i>transferable, professional skills are effectively demonstrated</i> <i>self management and independent learning are demonstrated</i>	professional skills Significant ability to learn independently and critically evaluate own progress using a wide range of feedback sources	transferable and professional skills Substantial ability to work independently and use feedback to reflect critically on own progress	professional skills Strong ability to work independently and use feedback to plan future tasks effectively	professional skills Evidence of developing well as an independent learner	professional skills Evidence of beginning to develop as an independent learner	professional skills Adequate evidence of beginning to develop as an independent learner	professional skills Limited evidence of ability to learn independently	professional skills Little or no evidence of ability to learn independently
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